

AdaCraft: Adaptive Educational Example Generation

Appendix

Human Evaluation A/B, Multilingual & Ablation Studies

Detailed Scenarios, Input/Output Examples, and Quantitative Results

Generated: May 2026

Source code: <https://github.com/anomalyco/AdaCraft>

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Appendix A: Human Evaluation & Multilingual Studies

A.1 Study A: Learner Self-Evaluation

Study A is a within-participant, self-report evaluation where 10 learners from Kolkata, India (Bengali cultural background) rated their own personalized examples. Each participant received 2 cold-start sessions (different topics), yielding 20 examples total. Participants scored the final (T3) example against 11 axes and compared baseline (T0) vs. final (T3) examples.

A.1.1 Evaluation Rubric (11 Axes)

Learners scored each final example on 5 standard axes (shared with Study B), 3 feedback-improvement axes, 2 self-report axes, and 1 bias axis:

Axis	Name	Scale	What is measured
PF	Personalization Fidelity	1-5	Does the example feel tailored to me specifically?
CC	Complexity Calibration	1-5	Was the depth/vocabulary right for my level?
CA	Conceptual Accuracy	1-5	Is the factual content correct?
PC	Pedagogical Clarity	1-5	Is it structured for learning?
DA	Domain Appropriateness	1-5	Are analogies/framing correct for this domain?
FC	Feedback Compliance	1-5	Did the final example reflect changes I asked for?
IS	Improvement Score	1-5	How much did T0 -> T3 improve for me?
IP	Informed Preference	T3/T0	Which example better helps me understand?
EU	Educational Usefulness	1-5	Did the final example help me understand better?
WU	Would Use	1-5	Would I share this with a peer from my background?
BF	Bias & Fairness	1-5	Free of stereotyping / reductive assumptions?

A.1.2 Participants (10 Raters)

All participants are from Kolkata, India with Bengali cultural background. Each is both the learner (whose profile generated the example) AND the evaluator. Sessions are cold-start with no prior system interaction.

- A1: Aritro Dutta (Engineering) A2: Ankita Chowdhury (Social Sciences)
- A3: Amritangshu Manna (Medicine) A4: Anindita Chatterjee (Education/B.Ed.)
- A5: Dr Subhobroto Dey (Educator/PhD) A6: Aritra Das (Mathematics)
- A7: Shrestha Banerjee (Economics) A8: Sreya Mondal (Nursing)
- A9: Sheila Chatterjee (BA) A10: Sourav (Educator/PhD)

A.1.3 Detailed Scenario Walkthroughs

Below are selected scenarios showing the full input/output/results pipeline for representative examples.

Scenario A.1.3a: EX01 -- Engineering Student, Natural Selection (DeepSeek)**Participant:** Aritro Dutta (A1) | Role: Engineering Student | Location: Kolkata, India**Topic:** Natural Selection | Provider: DeepSeek V3.2 | Start Mode: Cold**BASELINE EXAMPLE (T0) -- Generic beetle analogy**

Concept: Natural Selection

Example:

Consider a population of beetles living in a forest. Some beetles are brown, others are green. Birds (predators) hunt by sight. Green beetles blend into the leafy environment and are harder to spot, while brown beetles stand out against the green leaves and are eaten more frequently.

Over several generations:

1. Brown beetles are eaten more often and produce fewer offspring.
2. Green beetles survive longer, reproduce more, and pass on the green-colour trait.
3. Gradually, the beetle population shifts -- more and more beetles are green.

Key Insight:

Natural selection is not random. Traits that improve survival in a given environment become more common over time. The environment "selects" which traits persist -- not by intention, but by differential survival and reproduction.

FEEDBACK ROUND 1 (Input):

"The example is okay but feels too abstract. Can you make it more relatable to engineering work, like algorithm design or system optimisation?"

Agent Decision: regenerate

Concept: Natural Selection -- for an Engineering Student

Example:

Think of natural selection like an iterative optimisation algorithm -- similar to a genetic algorithm you might study in computational intelligence.

Imagine you're designing a load-bearing truss. You start with a population of 100 random designs. Each design is tested against the "fitness function": maximum load capacity at minimum material cost. The weakest designs (low fitness) are discarded. The strongest designs are selected, slightly mutated, and recombined to form the next generation.

This maps directly to biology:

Algorithm	Biology	
Random population	Genetic variation in a species	
Fitness function	Environmental survival pressure	
Selection	Differential survival & reproduction	
Mutation+crossover	Genetic mutation and sex. reprod.	

In Kolkata's engineering culture: Think of how IIT-JEE coaching institutes evolved. Only methods that produced results (high ranks) survived.

Key Insight:

Natural selection is nature's optimisation algorithm -- a massively parallelised search that has been running for billions of years.

FEEDBACK ROUND 2 (Input):

"Yes this is much better -- the genetic algorithm parallel really clicked for me. Can you add a concrete local example?"

Agent Decision: regenerate

... (The Kolkata coaching institute example was woven in as additional context) ...

Final Feedback: "Perfect, the coaching institute example from Kolkata makes it very tangible."**Agent Decision:** accept

	PF	CC	CA	PC	DA	Composite
T0 (Baseline)	2.31	3.87	4.94	4.61	3.72	4.12
T3 (Final)	4.10	4.20	5.00	4.30	4.00	4.45
Delta	+1.79	+0.33	+0.06	-0.31	+0.28	+0.33

FCR (Mean): 4.29 | Feedback Rounds: 3 | PF improvement: +1.79 points

Scenario A.1.3b: EX05 -- Medical Student, Natural Selection (DeepSeek)**Participant:** Amritangshu Manna (A3) | Role: Medical Student | Location: Kolkata, India**Topic:** Natural Selection | Provider: DeepSeek V3.2

The baseline T0 example was the same generic beetle analogy. The participant requested "something clinically relevant -- like how bacteria evolve resistance." The system regenerated using antimicrobial resistance (AMR) as the primary example, explicitly connecting MRSA, antibiotic stewardship, and selection pressure in clinical settings.

FINAL EXAMPLE (T3) Key Content:

Concept: Natural Selection -- for a Medical Student

The most clinically urgent example is antimicrobial resistance (AMR).

Scenario: A patient with *S. aureus* infection. You prescribe methicillin. Most bacteria die. A tiny fraction carry a mutation in the *mecA* gene that alters penicillin-binding proteins -> methicillin cannot bind -> they survive.

The drug acts as a SELECTION PRESSURE. Within days, the surviving resistant bacteria reproduce. The infection is now dominated by MRSA.

1. Variation -- random mutations create diversity
2. Selection -- antibiotic kills susceptible bacteria
3. Inheritance -- resistance genes passed to daughter cells (and via HGT)
4. Adaptation -- the population becomes resistant

Key Insight: Every time you prescribe an antibiotic, you are applying a selection pressure. Natural selection is happening inside your patients right now.

	PF	CC	CA	PC	DA	Composite
T0 (Baseline)	2.31	3.87	4.94	4.61	3.72	4.12
T3 (Final)	4.60	4.30	5.00	4.80	4.70	4.67
Delta	+2.29	+0.43	+0.06	+0.19	+0.98	+0.55

FCR (Mean): 4.33 | Rounds: 2 | PF improvement: +2.29 -- strongest PF gain in Study A**Scenario A.1.3c: EX08 -- Education Student, Plate Tectonics (GPT-5-nano)****Participant:** Anindita Chatterjee (A4) | Role: Education/B.Ed. Student**Topic:** Plate Tectonics | Provider: GPT-5-nano

This is one of only 2 examples where T0 was preferred over T3. The education student asked for a connection to education, but the system struggled to generate a satisfying domain-relevant personalisation within a single cold-start session.

	PF	CC	CA	PC	DA	Composite
T0 (Baseline)	2.27	3.76	4.91	4.48	4.12	4.14
T3 (Final)	2.80	3.50	4.60	3.60	3.40	3.73
Delta	+0.53	-0.26	-0.31	-0.88	-0.72	-0.41

FCR (Mean): 3.50 (lowest in study) | Rounds: 2 | Note: T3 composite DECLINED from T0

This failure case is instructive: Plate Tectonics is a highly geological topic with limited natural connections to education/pedagogy. The regeneration attempted a geopolitical metaphor but the participant found it 'still mostly geological.' This reveals a boundary condition for the cold-start context manager: when the topic domain and user domain have minimal semantic overlap, the system may struggle to generate a satisfying bridge example.

A.1.4 Study A -- Quantitative Results Summary

Axis	Mean	SD	Min	Max	Interpretation
PF	4.05	0.686	2	5	Personalization felt clearly tailored
CC	3.85	0.587	3	5	Complexity calibration adequate
CA	4.45	0.510	4	5	High conceptual accuracy (self-reported)
PC	4.20	0.616	3	5	Pedagogical structure solid
DA	4.00	0.725	3	5	Domain framing appropriate
FC	3.95	0.826	2	5	Most feedback was addressed
IS	4.15	1.040	2	5	Significant improvement ($p < 0.001$)
EU	3.95	0.605	3	5	Meaningful educational utility
WU	3.90	0.852	2	5	75% would share with a peer
BF	4.80	0.410	4	5	100% scored 4 or 5 -- no bias detected

Informed Preference (IP) -- T3 vs T0

T3 wins: 18/20 (90.0%) | T0 wins: 2/20 (EX08, EX17) | Binomial $p = 2.01e-4$ (significant)

Mean T3 vote share: 0.9. Both T0-preference examples involved Plate Tectonics for participants (education student, BA student) whose domain had minimal overlap with geology.

Statistical Tests

- IS t-test: mean=4.15 $t(19)=4.945$ $p=8.99e-5$ (IS > null=3.0)
- EU t-test: mean=3.95 $t(19)=7.025$ $p=1.09e-6$ (EU > null=3.0)
- WU t-test: mean=3.90 $t(19)=4.723$ $p=1.48e-4$ (WU > null=3.0)
- IP binomial: 18/20 T3 wins $p=2.01e-4$ (null=0.5)

Role Breakdown (PF by Role)

Role	n	Mean PF	Mean EU
economics_student	2	4.5	4.0
education_student	2	3.0	3.0
educator	4	4.0	4.25
engineering_student	2	4.5	4.0
mathematics_student	2	4.5	4.5
medical_student	2	4.5	4.5
nursing_student	2	4.0	4.0
social_sciences_student	2	4.0	4.0
undergraduate_student	2	3.5	3.0

Cross-Study PF Comparison

Study	PF Mean	Rater Type
Study A (self-report)	4.05	Learner self-report
Study B (expert-rated)	3.637	External domain experts (10 annotators)
Delta (A - B)	+0.413	Personalization more legible to recipients

A.2 Study B: Contextual Full-Rubric Evaluation

Study B is a between-participant expert evaluation where 10 annotators evaluated 30 examples (16 cold-start + 14 warm-start) across 4 professional roles. Each annotator scored each example on 7 axes (PF, CC, CA, PC, DA, FC, IS) using the same rubrics as Study A, plus an Informed Preference vote (T3 vs T0).

A.2.1 Annotators (10 Domain Experts)

- B1: Aritro Dutta (STEM)
- B2: Ankita Chowdhury (Social Sciences)
- B3: Amritangshu Manna (Life Sciences)
- B4: Anindita Chatterjee (Instructional Design)
- B5: Dr Subhobroto Dey (Generalist Educator)
- B6: Aritra Das (STEM)
- B7: Shrestha Banerjee (Social Sciences)
- B8: Sreya Mondal (Life Sciences)
- B9: Sheila Chatterjee (Instructional Design)
- B10: Sourav (Generalist Educator)

A.2.2 Results Summary (Human Mean Scores, n=30)

Axis	Mean	SD	Min	Max
PF	3.637	0.701	1.6	4.7
CC	3.867	0.582	2.4	5.0
CA	4.313	0.444	3.4	5.0
PC	3.940	0.564	2.6	5.0
DA	4.197	0.395	3.6	5.0
FC	3.707	0.578	2.4	4.6
IS	4.107	0.671	1.8	4.9

A.2.3 Informed Preference (T3 vs T0)

T3 wins: 26/30 examples (86.7%) | T0 wins: 4/30 (13.3%) | Mean T3 vote share: 85.3%

Binomial test: $p = 2.97e-5$ (one-sided, null=0.5). Highly significant.

T0-majority examples: EX09, EX11, EX20, EX28.

A.2.4 Inter-Rater Reliability -- ICC(2,1)

Axis	ICC	95% CI	F	Interpretation
PF	0.692	[0.612, 0.832]	27.36	Moderate
CC	0.596	[0.479, 0.748]	16.67	Moderate
CA	0.473	[0.340, 0.637]	10.04	Fair
PC	0.540	[0.484, 0.752]	16.96	Moderate
DA	0.465	[0.359, 0.654]	10.78	Fair
FC	0.566	[0.516, 0.773]	19.01	Moderate
IS	0.684	[0.617, 0.835]	27.93	Moderate

All ICCs significant at $p < 0.001$. CA and DA are 'Fair' (< 0.5), reflecting legitimate domain-level variation.

A.2.5 Human--LLM Judge Correlation (Pearson r)

Axis	r	p	n
PF	0.820	2.89e-8 ***	30
CC	0.728	5.25e-6 ***	30
CA	N/A (zero var.)	--	30
PC	0.696	1.93e-5 ***	30
DA	0.025	0.896 (n.s.)	30
Composite	0.629	0.0002 ***	30
FC vs FCR	0.748	2.03e-6 ***	30

Strong human-LLM alignment on PF ($r=0.820$), CC ($r=0.728$), and PC ($r=0.696$). DA shows no correlation ($r=0.025$, n.s.) -- likely

because LLM judge scores cluster near ceiling with minimal variance while expert human raters show broader spread. FC vs FCR correlation ($r=0.748$) validates the LLM-based feedback compliance metric.

A.2.6 Cold-Start vs Warm-Start Comparison

Condition	n	Mean PF	Mean IS
Cold	16	3.531	3.894
Warm	14	3.757	4.350
Delta (Warm - Cold)	--	+0.226	+0.456

Profile Breakdown -- Mean PF by Role

Role	n	Mean PF	SD
Engineer	7	4.057	0.597
Student	8	3.688	0.601
Humanities Researcher	7	3.457	1.061
Nurse	8	3.375	0.337

A.3 Multilingual Evaluation

The multilingual evaluation tests AdaCraft's ability to generate and adapt personalized educational examples across 6 languages (Hindi, Tamil, Bengali, German, Arabic, Mandarin) under 3 scenarios: FNL (Full Native Language), NFEO (Native Feedback, English Output), and EN_BASELINE (all English). Three providers tested: GPT-5-nano, DeepSeek V3.2, Sarvam (Indic only).

A.3.1 Experimental Design

Scenario	Topic Lang.	Feedback Lang.	Output Lang.	Purpose
FNL	Native (X)	Native (X)	Native (X)	End-to-end multilingual pipeline
NFEO	English	Native (X)	English	Cross-lingual feedback processing
EN_BASELINE	English	English	English	Comparison baseline

Languages and Profiles

User ID	Name	Language	Code	Role	Location
ml_user_01	Priya	Hindi	hi	Student	Mumbai, India
ml_user_02	Meena	Tamil	ta	Nurse	Chennai, India
ml_user_03	Arjun	Bengali	bn	Software Engineer	Kolkata, India
ml_user_04	Klaus	German	de	Humanities Researcher	Munich, Germany
ml_user_05	Fatima	Arabic	ar	Nurse	Cairo, Egypt
ml_user_06	Wei	Mandarin	zh	Software Engineer	Shanghai, China

Test Matrix

Scenario	Profiles	Topics	Cells (GPT/DeepSeek)	Cells (Sarvam)
FNL	6	4	24	12 (Indic only)
NFEO	6	4	24	12 (Indic only)
EN_BASELINE	6	4	24	12 (Indic only)
Total	--	--	72	36

A.3.2 GPT-5-nano Results

Scenario	Group	n	Composite	LangMatch	FCR@3	FCR@4	LUR	DA
FNL	Indic	12	4.800	1.000	0.917	0.708	1.000	0.667
FNL	Foreign	12	4.775	0.500	0.792	0.708	1.000	0.667
FNL	Overall	24	4.788	0.800	0.854	0.708	1.000	0.667
NFEO	Indic	12	4.800	0.917	0.833	0.792	1.000	0.667
NFEO	Foreign	12	4.833	0.833	0.958	0.792	0.917	0.639
NFEO	Overall	24	4.817	0.875	0.896	0.792	0.958	0.653
EN_BASE	Overall	24	4.758	--	--	--	--	--

Mandarin LangMatch = 0.000 across all 4 FNL cells (both providers). This is a cross-provider systematic issue: the model generates English output despite Chinese topic/feedback, likely because the prompt lacks an explicit language enforcement directive.

A.3.3 DeepSeek V3.2 Results

Scenario	Group	n	Composite	LangMatch	FCR@3	FCR@4	LUR	DA
FNL	Indic	12	4.775	1.000	1.000	0.909	0.917	0.944
FNL	Foreign	12	4.808	0.500	0.864	0.682	1.000	0.819
FNL	Overall	24	4.792	0.800	0.932	0.795	0.958	0.882
NFEO	Indic	12	4.817	0.917	0.917	0.500	1.000	0.778
NFEO	Foreign	12	4.817	1.000	0.917	0.750	1.000	0.833
NFEO	Overall	24	4.817	0.958	0.917	0.625	1.000	0.806
EN_BASE	Overall	24	4.817	--	--	--	--	--

A.3.4 Sarvam Results (Indic-only)

Scenario	Group	n	Composite	LangMatch	FCR@3	FCR@4	LUR	DA
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FNL	Indic	12	4.642	1.000	0.792	0.542	0.917	0.667
NFEO	Indic	12	4.767	0.917	0.667	0.458	1.000	0.667
EN_BASE	Indic	12	4.725	--	--	--	--	--

A.3.5 Wilcoxon Test Results (All 10 tests -- None Significant)

Provider	Scenario	Group	n	Delta	p-val	d	Sig (a'=0.025)
GPT-5-nano	FNL	Indic	12	-0.017	0.655	-0.130	No
GPT-5-nano	FNL	Foreign	12	+0.075	0.547	+0.158	No
GPT-5-nano	NFEO	Indic	12	-0.017	0.655	-0.130	No
GPT-5-nano	NFEO	Foreign	12	+0.133	0.317	+0.331	No
DeepSeek	FNL	Indic	12	-0.058	0.317	-0.316	No
DeepSeek	FNL	Foreign	12	+0.008	0.705	+0.080	No
DeepSeek	NFEO	Indic	12	-0.017	0.739	-0.097	No
DeepSeek	NFEO	Foreign	12	+0.017	0.564	+0.169	No
Sarvam	FNL	Indic	12	-0.083	0.389	-0.351	No
Sarvam	NFEO	Indic	12	+0.042	0.458	+0.195	No

Interpretation: No provider shows a statistically significant quality change under either multilingual scenario vs the English baseline (all $p > 0.025$ Bonferroni threshold). Effect sizes are all small ($|d| < 0.36$). The null finding is consistent across both providers and both language groups -- multilingual interaction does not degrade (or inflate) composite example quality.

A.3.6 Key Findings

Finding 1: No Quality Degradation from Multilingual Interaction

All 10 Wilcoxon tests non-significant ($p > 0.30$). Composite scores under FNL and NFEO statistically indistinguishable from English baseline. The AdaCraft pipeline generalizes to multilingual input/output without measurable quality cost.

Finding 2: Mandarin LangMatch Failure is Cross-Provider

FNL LangMatch for Mandarin (zh) = 0.000 across both GPT-5-nano and DeepSeek (8/8 cells). Arabic, Hindi, Tamil, Bengali achieve perfect LangMatch. German excluded (Latin-script ambiguity).

Finding 3: DeepSeek Superior Decision Accuracy

DeepSeek DA (0.944 FNL Indic) substantially exceeds GPT-5-nano (0.667) and Sarvam (0.667). The 0.667 floor represents correct R2/R3 (accept, flag_pattern) but failure on R1 (regenerate).

Finding 4: Sarvam Lower Composite and FCR vs. General-Purpose LLMs

Sarvam FNL composite (4.642) is 0.150-0.158 below GPT-5-nano and DeepSeek. Tamil weakest per-language cell (4.375) with FCR@3=0.500. However LangMatch=1.000 across all Indic cells.

Appendix B: Main Ablation Study

The main ablation study evaluates the AdaCraft pipeline using a 4-tier ablation design (T0/T1/T2/T3) across 8 synthetic users (4 profiles x 2 start modes: cold/warm), 4 topics, and 2 LLM providers (GPT-5-nano, DeepSeek V3.2). Total: 128 cells per provider.

B.1 Experimental Design

B.1.1 4-Tier Architecture

Tier	Profile	Context Mgr	Feedback	Description
T0	Skipped	Skipped	Skipped	Generic LLM -- no personalization
T1	Injected	Skipped	Skipped	Static profile only
T2	Injected	Active	Skipped	Profile + learning history context
T3	Injected	Active	Active	Full AdaCraft system

B.1.2 Synthetic Users (4 Profiles x 2 Start Modes)

User ID	Name	Role	Location	Start	R1 Feedback
eval_user_01	Lena	Student	Berlin, Germany	cold	F1: "too complicated"
eval_user_02	Amara	Nurse	Lagos, Nigeria	cold	F2: "not my field"
eval_user_03	Carlos	Humanities Researcher	Sao Paulo, Brazil	cold	F1: "too complicated"
eval_user_04	Kenji	Software Engineer	Tokyo, Japan	cold	F2: "not my field"
eval_user_05	Lena	Student	Berlin, Germany	warm	A1: "I dont get it"
eval_user_06	Amara	Nurse	Lagos, Nigeria	warm	A2: contradictory
eval_user_07	Carlos	Humanities Researcher	Sao Paulo, Brazil	warm	A5: hidden critique
eval_user_08	Kenji	Software Engineer	Tokyo, Japan	warm	A4: "Hmm."

4 Topics: Natural Selection (evolutionary_biology), Cognitive Bias (psychology), Compound Interest (mathematics_finance), Plate Tectonics (earth_sciences). All 4 topics crossed with all 8 users = 32 cells per tier.

B.1.3 Scripted Feedback Battery (T3 only)

Round	Key	Feedback Content	Expected Decision
R1 (cold)	F1/F2	Unambiguous complaints: "too complicated" or "not my field"	regenerate
R1 (warm)	A1/A2/A4/A5	Adversarial: vague, contradictory, minimal, hidden critique	regenerate / accept
R2 (all)	F3	"This is great! The example really helped me understand." (positive close)	accept
R3 (all)	FP	Stable-trait preference per role (student/nurse/researcher/engineer)	flag_pattern

The adversarial battery tests the Adaptive Response Agent's robustness to realistic, ambiguous user input. A1 = vague, A2 = contradictory ("make it simpler but go deeper"), A4 = minimal ("Hmm."), A5 = hidden critique with surface praise.

B.1.4 Metrics

- Composite Score = $0.20 \cdot PF + 0.20 \cdot CC + 0.30 \cdot CA + 0.20 \cdot PC + 0.10 \cdot DA$ (1-5 scale)
- FCR@3 / FCR@4: Fraction of regenerations with compliance score ≥ 3 / ≥ 4
- LUR: Fraction of T3 sessions with at least 1 regeneration
- PPU: PF delta between warm-start T3 and warm-start T1
- Decision Accuracy: Fraction of agent actions matching ground-truth expected decisions

B.2 Ablation Results

B.2.1 Composite Scores by Tier

Tier	DeepSeek V3.2	GPT-5-nano	Description
T0	4.257 +/- 0.057	3.712 +/- 0.280	Generic -- no profile, no context
T1	4.809 +/- 0.037	4.388 +/- 0.261	+ User profile only
T2	4.859 +/- 0.025	4.791 +/- 0.147	+ Context instruction from learning history
T3	4.888 +/- 0.015	4.981 +/- 0.058	+ Feedback loop (full system)
T0->T1 Delta	+0.552	+0.676	
T0->T3 Delta	+0.631	+1.269	

DeepSeek starts from a higher baseline (T0=4.257 vs 3.712) suggesting stronger zero-shot personalization. GPT-5-nano shows a larger T0->T3 gain (+1.269 vs +0.631). Both reach comparable T3 ceilings (~4.89 vs ~4.98).

B.2.2 Per-Axis Breakdown

DeepSeek V3.2

Axis	T0	T1	T2	T3
PF (Personalization Fidelity)	2.73	4.80	4.93	4.99
CC (Complexity Calibration)	4.13	4.35	4.43	4.47
CA (Conceptual Accuracy)	4.94	4.99	5.00	5.00
PC (Pedagogical Clarity)	4.82	4.93	4.95	4.99
DA (Domain Appropriateness)	4.39	4.95	4.99	5.00

GPT-5-nano

Axis	T0	T1	T2	T3
PF (Personalization Fidelity)	2.09	3.47	4.34	4.94
CC (Complexity Calibration)	3.78	4.38	4.84	5.00
CA (Conceptual Accuracy)	4.41	4.78	4.97	5.00
PC (Pedagogical Clarity)	4.00	4.53	4.88	4.97
DA (Domain Appropriateness)	4.16	4.78	4.88	5.00

PF shows the largest tier-to-tier jump for both providers. For GPT-5-nano, PF rises from 2.09 (T0, essentially generic) to 4.94 (T3, near-perfect personalization) -- a +2.85 point gain. This validates the core hypothesis: the combination of profile injection, context management, and adaptive feedback produces dramatically more personalized examples.

B.3 Statistical Significance

B.3.1 Friedman Test (Omnibus, all 4 tiers)

Provider	Chi-squared	p-value	Significant
DeepSeek V3.2	96.000	1.13e-20	***
GPT-5-nano	91.351	1.12e-19	***

B.3.2 Pairwise Wilcoxon Signed-Rank (Holm-Bonferroni corrected, 6 pairs)

Pair	DeepSeek Diff	DS r_rb	DS p(Holm)	GPT-5 Diff	GPT r_rb	GPT p(Holm)
T0 vs T1	+0.552	1.000	2.79e-09 ***	+0.675	1.000	2.79e-09 ***
T0 vs T2	+0.602	1.000	2.79e-09 ***	+1.078	1.000	2.79e-09 ***
T0 vs T3	+0.631	1.000	2.79e-09 ***	+1.269	1.000	2.79e-09 ***
T1 vs T2	+0.049	1.000	2.79e-09 ***	+0.403	1.000	6.95e-06 ***
T1 vs T3	+0.078	1.000	2.79e-09 ***	+0.594	1.000	6.95e-06 ***
T2 vs T3	+0.029	1.000	2.79e-09 ***	+0.191	1.000	1.22e-05 ***

All 6 pairwise comparisons are significant for both providers. The rank-biserial $r = 1.000$ for all pairs means ALL 32 user-topic pairs improved monotonically -- zero exceptions. This is an extremely large and consistent effect.

B.3.3 Interpretation

The tier improvements are not just statistically significant but practically meaningful. The monotonic improvement across all 32 cells ($r_{rb} = 1.0$) confirms that adding each successive capability layer (profile, context, feedback) reliably improves example quality. The largest jump is T0->T1 (profile injection), particularly for PF. The T2->T3 jump (feedback loop) is smaller but still significant, suggesting that the largest gains come from personalization infrastructure rather than the feedback mechanism itself.

B.4 Convergence Metrics

B.4.1 Feedback Compliance Rate (FCR)

Battery	DeepSeek FCR@3	DS FCR@4	DS Mean	GPT-5 FCR@3	GPT FCR@4	GPT Mean
Easy	1.000	1.000	4.536	0.925	0.850	4.475
Adversarial	1.000	0.692	3.923	1.000	1.000	4.800
Overall	1.000	0.902	4.341	0.945	0.891	4.564

DeepSeek achieves perfect FCR@3 (1.000) overall; GPT-5-nano is close (0.945). Both providers show strong feedback compliance. DeepSeek's FCR@4 on adversarial battery (0.692) is notably lower than GPT-5-nano's perfect score (1.000), suggesting GPT-5-nano handles vague/contradictory feedback more reliably at higher compliance thresholds.

B.4.2 Loop Utilization Rate (LUR)

Battery	DeepSeek LUR	DS Triggered	GPT-5 LUR	GPT Triggered
Easy (cold)	1.000	16/16	1.000	16/16
Adversarial (warm)	0.875	14/16	0.938	15/16
Overall	0.938	30/32	0.969	31/32

Both providers trigger regeneration at very high rates ($\geq 93.8\%$). Cold-start users always trigger regeneration (100%), while 2-3 warm-start adversarial cases are accepted without regeneration -- the agent judges the feedback does not require a change.

B.4.3 Pattern Persistence Utilization (PPU)

Group	DeepSeek Mean PF	GPT-5-nano Mean PF
Warm T3 (full system)	5.000	5.000
Warm T1 (no patterns, no feedback)	4.688	3.062
Cold T3 (no prior history)	4.979	4.875
Delta PF (Warm T3 - Warm T1)	+0.312	+1.938

PPU measures whether stored learning patterns improve personalization at generation time. GPT-5-nano shows a dramatic PPU delta (+1.938) because its warm-start T1 PF is notably low (3.062). When patterns are available at generation time (warm T3), PF jumps to 5.000 -- the system uses stored domain_preference patterns to insert appropriate domain context. DeepSeek's smaller delta (+0.312) reflects its already-strong warm T1 PF (4.688), leaving less room for improvement.

B.5 Detailed Scenario Walkthroughs

B.5.1 T0 Scenario -- Lena, Natural Selection (GPT-5-nano)

Tier T0 bypasses all personalization layers. The example uses a generic beetle analogy with no reference to Lena's profile (Berlin student). Feedback is skipped (eval_mode gate).

T0 GENERATED EXAMPLE (Initial + Final are identical -- no feedback processing):

Concept: Natural Selection

Example:

Think of a population of beetles living on a tree trunk. The beetles vary in color: some are light, some are dark. In this environment, birds tend to spot the light-colored beetles against the dark bark, so light beetles are more likely to be eaten. If the bark is dark, dark beetles blend in better and are more likely to survive and have babies.

Starting numbers this season:

- 40 light beetles
- 60 dark beetles

Predation and survival:

- Light beetles survive at 20% -> 8 survive
- Dark beetles survive at 60% -> 36 survive

Result: Next generation: ~8 light offspring, ~36 dark offspring.

Dark-colored beetles become more common over time.

Axis	PF	CC	CA	PC	DA	Composite
T0 Score	2	4	4	3	3	3.3

PF=2: 'Completely generic -- could have been written for anyone.' No location, no role adaptation. The beetle analogy is a standard textbook example. CA=4 is adequate (factually correct).

B.5.2 T3 Scenario -- Lena, Natural Selection (GPT-5-nano)

Tier T3 runs the full pipeline: profile injection -> context manager -> generation -> 3 feedback rounds (R1: simplify, R2: accept, R3: school-life scenario).

ROUND 0 (Initial Generation -- already personalized):

Concept: Natural Selection

In a temperate European forest near Berlin, a population of moths shows color variation: light and dark forms. The trait is heritable, and color affects how easily predators spot moths on tree trunks.

Setup: 100 moths (70 light, 30 dark)

Environment: Tree bark becomes darker

Survival rates: light 0.20, dark 0.60

Surviving: light 14, dark 18 -> each leaves 2 offspring

Next gen: 28 light, 36 dark -> dark = 56.3%

Insight: The dark form becomes more common because it has higher fitness.

Axis	PF	CC	CA	PC	DA	Composite
Round 0 Score	5	5	5	5	5	5.0

PF=5: Already references Berlin, European forest. Complexity appropriate for high school student. The initial T3 example scores perfectly on all axes -- the profile injection alone achieves strong personalization for this user-topic pair.

ROUND 1 -- Feedback + Regeneration:

User: "This is too complicated for me. Can you simplify it with a more basic example?"

Agent Action: regenerate

Light survivors=1, Dark survivors=3. Each produces 1 offspring.

Next gen: 2 light, 3 dark (dark = 60%).

FCR compliance score: 4/5 -- simplified correctly but still slightly numerical for a beginner request.

ROUND 2 -- Positive Close:

User: "This is great! The example really helped me understand the concept."

Agent Action: accept

User: "I always connect better when examples use everyday scenarios from school life, like assignments or exams."

Agent Action: regenerate

In Lena's biology class in Berlin, 20 students = mini-population.
Alleles: A (preparedness), a (lower). AA=9, Aa=6, aa=5.
Hard exam -> selection pressure. w_AA=1.0, w_Aa=0.9, w_aa=0.4.
After selection: p(A) changes 0.60 -> 0.714.
Key Insight: Selection pressure shifts allele frequencies toward advantageous traits.

Axis	PF	CC	CA	PC	DA	Composite
Final Score	4	5	5	4	4	4.5

Final PF=4 (vs initial 5): the classroom scenario is less location-specific than the Berlin forest reference. But the example now explicitly integrates the user's stated preference for school-life scenarios -- demonstrating the adaptive response loop.

B.5.3 Decision Accuracy

Provider	Battery	Correct	Total	Accuracy
DeepSeek V3.2	Overall T3	--	--	0.926
GPT-5-nano	Overall T3	2.0 avg	3 rounds	0.667

GPT-5-nano consistently achieves 0.667 accuracy: it correctly identifies R2 (accept) and R3 (flag_pattern) but frequently fails R1 (regenerate) -- instead accepting the complaint and providing an insight. DeepSeek's higher accuracy reflects better sensitivity to domain-complaint signals (F2 feedback type).

B.6 Inter-Judge Agreement & Cross-Provider Summary

B.6.1 Cohen's Kappa (Primary: GPT-4.1-nano / Secondary: Llama 3.3 70B)

Metric	DeepSeek V3.2	GPT-5-nano
Subsampled cells	26	26
Axis-level pairs	130	130
Exact agreement	107 (82.3%)	107 (82.3%)
Off by 1	13 (10.0%)	22 (16.9%)
Off by 2+	10 (7.7%)	1 (0.8%)
Cohen's kappa	0.607 (Substantial)	0.681 (Substantial)

Both providers show 'Substantial' agreement (Landis & Koch) between the primary and secondary LLM judges. GPT-5-nano has fewer severe disagreements (0.8% off by 2+ vs DeepSeek 7.7%).

B.6.2 Cross-Provider Summary Comparison

Metric	DeepSeek V3.2	GPT-5-nano	Interpretation
T0 composite	4.257	3.712	DeepSeek stronger baseline
T3 composite	4.888	4.981	GPT-5-nano converges higher
T0->T3 gain	+0.631	+1.269	GPT-5-nano benefits more from full pipeline
FCR@4 overall	0.902	0.891	Comparable compliance
LUR overall	0.938	0.969	Both high loop utilization
PPU delta PF	+0.312	+1.938	GPT-5-nano shows larger pattern benefit
Cohen's kappa	0.607	0.681	Both 'Substantial' agreement
Decision Accuracy	0.926	0.667	DeepSeek better on regenerate decisions
Friedman chi2	96.000	91.351	Both highly significant

B.6.3 Key Takeaway

The ablation study conclusively demonstrates that the AdaCraft pipeline produces monotonically improving example quality across all 4 tiers, with statistically significant gains at every layer (profile injection, context management, adaptive feedback). GPT-5-nano shows larger relative gains (+1.269 T0->T3) while DeepSeek V3.2 has a stronger zero-shot baseline (4.257). Both providers reach comparable T3 performance (~4.9-5.0), suggesting an empirical ceiling for the 5-axis G-Eval rubric.